Spectral analyser is widely used in the first step of the feature detection process. The spectrogram can be useful in gathering information about a signal and in determining which features can be extracted from a signal. The spectrogram is used for defining statistical features of the signal, such as mean value, standard deviation, skewness and kurtosis. These features build a feature vector to retrieve similar signal from the database. Some of the other features that can be extracted are peak to peak value, power, energy, entropy, total harmonic distortion and signal to noise ratio. The process of feature extraction using wavelets is explained in [21].

Feature extraction process is initiated after the level of decomposition and reconstruction is performed with the wavelet coefficient. During the process of feature extraction critical information from the signals are obtained which aids in classification of different fault states [36]. The process of feature extraction is usually performed using spectral analyser. The spectrogram aids in attaining information regarding signal and identification of feature that can be extracted. Statistical feature such as mean value, standard deviation, skewness and kurtosis are defined using spectrogram. A feature vector is formed by the features for retrieving similar signals from the database. Feature that can be extracted are peak to peak value, power, energy, entropy, total harmonic distortion and signal to noise ratio. These features can be expressed as follows:

Mean of the reconstructed signal is calculated by

$$\text{Mean}\ (\mu) = \frac{1}{N}\sum_{i=0}^{N-1} V_i \tag{8.19}$$

where N denotes the number of samples and V_i is the sampled reconstructed signal.

Standard deviation of the reconstructed signal is calculated by

$$\text{Standard deviation}\ (\sigma) = \frac{1}{N-1}\sum_{i-0}^{N-1}(V_i-\mu)^2 \tag{8.20}$$

Skewness and kurtosis of the reconstructed signal is given by

$$\text{Skewness}\ (S) = \frac{\frac{1}{N}\sum_{i=0}^{N}(V_i-\mu)^3}{\left(\sqrt{\frac{1}{N}\sum_{i=0}^{N}(V_i-\mu)^2}\right)^3} \tag{8.21}$$

$$\text{Kurtosis}\ (K) = \frac{\frac{1}{N}\sum_{i=0}^{N}(V_i-\mu)^4}{\left(\sqrt{\frac{1}{N}\sum_{i=0}^{N}(V_i-\mu)^2}\right)^4} \tag{8.22}$$

Peak to peak value of the reconstructed signal is given by

$$V_{pp} = 2\sqrt{2}\sigma \tag{8.23}$$

Energy of the decomposed signal is formulated using

$$E = \int_{-\infty}^{\infty} |V(t)^2| dt \tag{8.24}$$

Power of the reconstructed signal is given by

$$P = \lim_{N \to \infty} \frac{1}{2T} \oint_{-T}^{T} \left| V(t)^2 \right| dt \tag{8.25}$$

where $V(t)$ is the reconstructed signal.

Entropy is defined as a major tool in information theory. It is also used to estimate the type of wavelet suitable for decomposing and reconstructing a given signal. The entropy of a given signal is found by

$$H(V) = -\sum_{i=1}^{N} p(V_i) \log_{10} p(Vi) \tag{8.26}$$

where $p(V_i)$ is given by probability of sample of voltage signal.

To estimate the total harmonic distortion of a sinusoidal signal in time domain, we use (8.28):

$$\begin{aligned} y(t) &= \alpha_0 + \frac{\alpha_2}{2} + \frac{3\alpha_4}{8} + \left(\alpha_1 + \frac{3\alpha_3}{4} + \frac{10\alpha_5}{16}\right) \\ &\quad sin\omega_0 t - \left(\frac{\alpha_2}{2} + \frac{\alpha_4}{2}\right) cos2\omega_0 t - \left(\frac{\alpha_3}{4} + \frac{5\alpha_5}{16}\right) \\ &\quad sin3\omega_0 t + \frac{\alpha_4}{8} cos4\omega_0 t + \frac{\alpha_5}{16} sin5\omega_0 t \end{aligned} \tag{8.27}$$

where α_i =coefficients of Taylor series.

The signal to noise ratio for a given signal is determined by the ratio of reconstructed signal to original signal:

$$\text{SRN} = \frac{\text{Reconstructed signal}}{\text{Original signal}} \tag{8.28}$$

Once the required features of all the faults and operating conditions of PV systems were extracted, we apply principal component analysis (PCA) to minimize the feature set.

8.4.4 Principle component analysis

PCA is a statistical analysis tool that utilizes multiple dimensions data set for minimization and highlighting the similarity within data set. In case of high dimension data, the similarity identification is very difficult hence for such conditions the PCA analyses the data graphically. After the similarity are identified then the data is reduced with insignificant loss of information. These dimensions reduction can be achieved by transforming the original data set into a series of uncorrelated principal components. Principle algorithm for PCA is shown in Figure 8.7.

By applying PCA, the features are minimized into three uncorrelated variables which were in turn utilized with the ML techniques to obtain the trained data for fault classification.

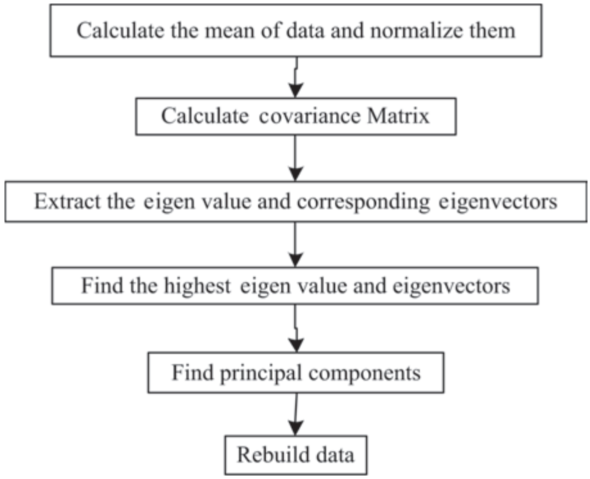


Figure 8.7 Diagram flow of the PCA algorithm

8.5 Machine learning approach

ML is a cluster of algorithms that are used to obtain models from data. There are two main types of ML models. First is the supervised learning model and second is the unsupervised learning model. In this research, the supervised ML model is adapted through K-NNs for the classification process [85].

8.5.1 K-nearest neighbour classifier

The K-NN is a classification learner method which is used for categorizing data into different classes. An assumption in matric space Ω regarding set of points is considered and each point is assigned a label of either 0 or 1. The labelled samples are represented by $(X_1, Y_1), (X_2, Y_2), \ldots (X_n, Y_n)$ and (X, Y) are in query. The label of query is predicted based on the most common class among the k closest point to the labelled sample X as illustrated in the Figure 8.8.

Case of ties has been avoided in the considered example. The figure illustrates two possible cases where ties can occur in algorithm. It is feasible to having multiple classes transpiring in identically frequently among the query of K-NNs. Even equal distance tie from the multiple points in query is possible. Distribution with density has also been discussed in may research for avoiding distance tie. Random selection is also one of the common methods for breaking ties, hence if a voting tie occurs then a random tie is selected from the most common labels and in case of a distance tie, a random point is selected at a distance. To avoid issues of binary classification, the vote tie is broken by selecting the label 1. By generating random variable $U_1, U_2, \ldots, U_n$, distance tie is broken in case of uniform distribution on [0,1]. In case of distance tie between two point X_i and X_j, X_i is selected if $U_i > U_j$ and X_j is selected if $U_j > U_i$. The pseudocode is presented in algorithm as follows.

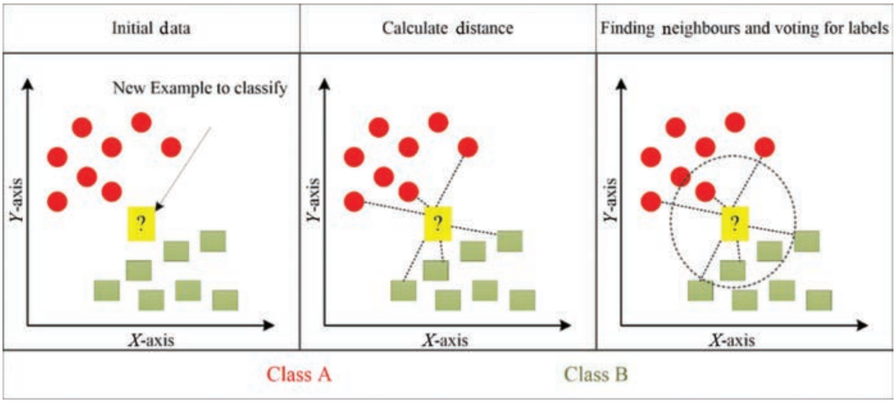


Figure 8.8 Illustration of K-NN

Algorithm: *k*-NN pseudocode

Require: $k \epsilon N$, X is a domain, Y is the response (must be a finite set$\{1,2,\ldots,p\}$),

$\alpha \epsilon X\left(x_1, y_1\right), \ldots., \left(x_n, y_n\right) \epsilon X \times Y$

{Calculate input point distance from all the data points}
for $i = 1$ **to** n **do**
$d_i \leftarrow d\,(a, x_i)$
end for
{Finding *k*-nearest neighbours response for the input point}
for $i = 1$ **to** k **do**
$m \leftarrow \arg_m \min\,\{d_m$ such that $1 \leq m \leq n$ not previously selected}

$a_i \leftarrow y_m$

end for
{Finding the number of iterations}
for $i = 1$ **to** p **do**
$d_i \leftarrow d(a, X_i)$
end for
$r \leftarrow \{y_i\, |1 \leq i \leq p$ such that v_i is maximum among $v_1, v_2, \ldots, v_p\}$
{identification of most common response among *k* nearest neighbours, and in case of multiple responses pick a fixed one}
return *r* {retuning the common response}

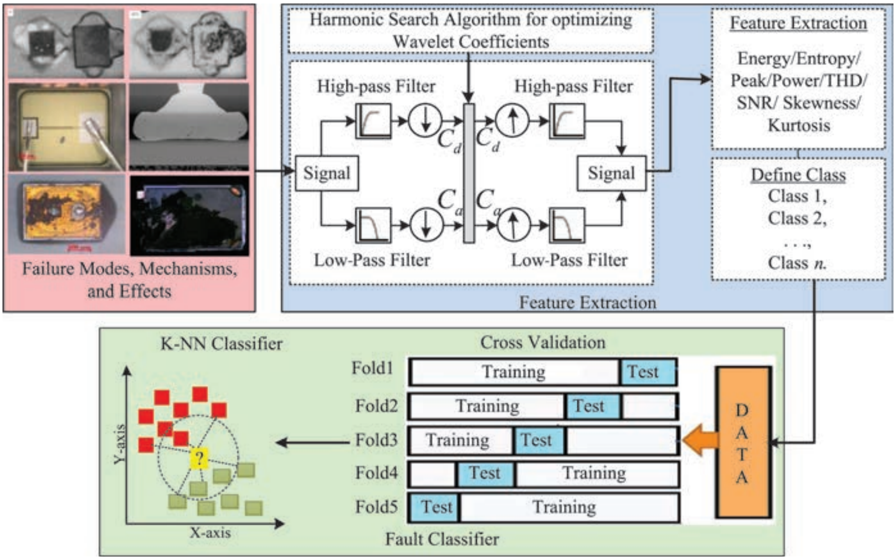
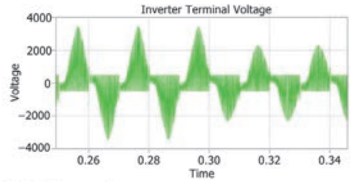


Figure 8.9 Block diagram for the wavelet-based fault detection technique.

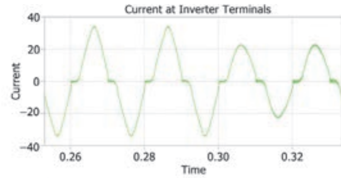
8.5.2 Fault classification algorithm

The methodology adapted for fault classification is structured as depicted in Figure 8.9. In this research, the fault classification technique is used to learn a model called a classifier. Data corresponding to various faults and operating conditions of PV system is divided into two groups: training set and testing set. In the training phase, the training set is fed to the classifier for labelled data set into one of the classes depending on the target output. A fivefold cross validation is applied to validate the trained data. In the testing phase, the test samples are verified depending on the target output. Once essential features have been identified, the classification of a fault condition is straightforward performed. The K-NN technique discussed in Section 8.5.1 is used for fault classification due to its advantages with nonlinear data classification as mentioned in Section 8.5.1. For simulation analysis, the input data (i.e., the extracted features) are tabulated and imported to the classification program. The classifier models are trained for functional fault classification.

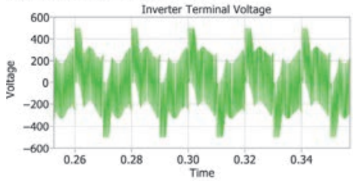
To observe the performance of the proposed methodology, a 4-kW two-stage PV system for a fixed load is simulated for a varying climatic and load conditions. For experimental purpose, six different modes of operation (normal, bond-wire failure, solder fatigue, overstress due to ESD, wear out due to substrate cracking and other failure conditions) are categorized for the operation of the power PV inverter. The terminal voltage and current measurements of the inverter under different modes of operation for a determined time period are logged. A sample of waveforms recorded for all the operating conditions and failure mechanisms for developing the fault classification algorithm are depicted in Figure 8.10. The process of feature vector representation and applying DWT through HSA for the purpose of feature extraction



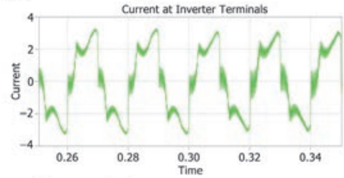
(e) *Voltage during overstress*



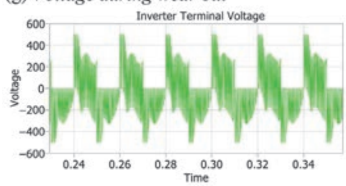
(f) *Current during overstress*



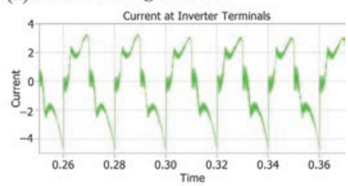
(g) *Voltage during wear out*



(h) *Current during wear out*



(i) *Voltage during other failures*



(j) *Current during other failures*

Figure 8.10 Various operating and fault conditions of PV inverter

is carried out. Eight different features such as energy, entropy, power, peaks, harmonics, signal to noise ratio, skewness and kurtosis are extracted for voltage and current outputs of each condition. This forms a feature set matrix 4 013 × 8, where the normal operation has 513 × 8 features, bond-wire failure has 1 448 × 8 features, solder fatigue, overstress, wear out and other failures have 513 × 8 features each.

The parameters of the classifier for the training process and the corresponding outcomes are shown in Table 8.5.

The classification accuracy is defined as the ratio of correctly classified samples to the total number of samples in the test data set. The corresponding equation is given as

Table 8.5 Model type and training performance results for K-NN

Model type	
Number of neighbours	10
Distance metric	Euclidean
Distance weight	Squared inverse
Standardize data	True
Results	
Accuracy	96.1%
Total misclassification cost	157
Prediction speed	~22 000 obs/s
Training time	4.5238 s

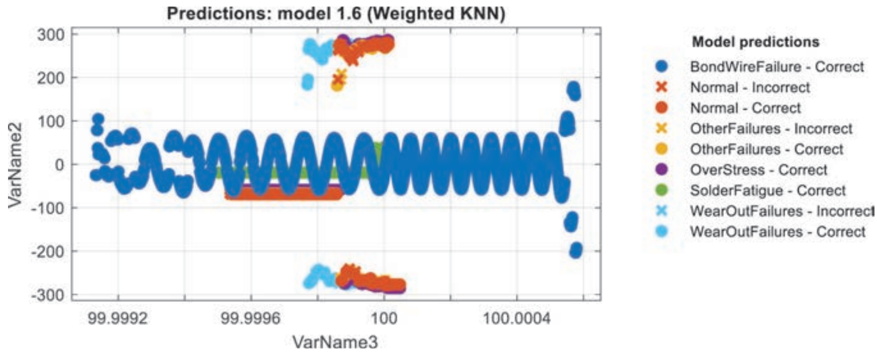


Figure 8.11 Scatter plot of the trained data

$$\text{Accuracy of classification} = \frac{\text{Total no. of samples classified correctly}}{\text{Total no. of samples in the data set}} \times 100 \quad (8.29)$$

To confirm the developed classifier data, a set of input data not used within the training stage, is given to the trained data. This technique improves the average accuracy about 96.1 per cent in training process. The corresponding results were depicted in Figures 8.11–8.13.

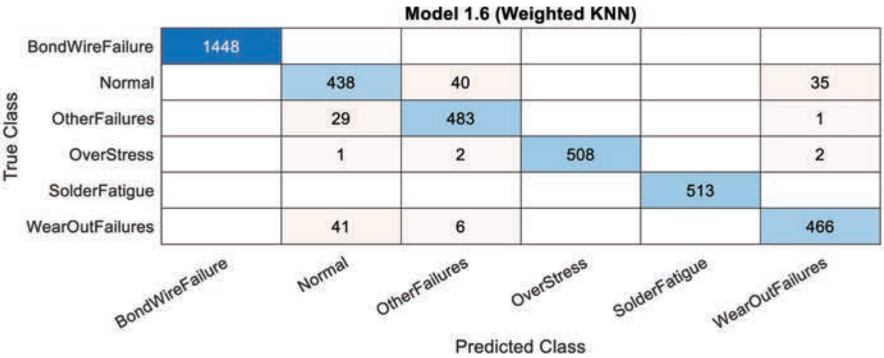
Figure 8.11 displays the scatter plot between two features of the training data. The scatter plot, explores the data for important predictors, outliners and visual patterns.

Figure 8.12 depicts the confusion matrix for the experiment. The confusion matrix aids in understanding the performance of the classifier in each of the classes. It supports the understanding that, if the classifier is executed poorly in identifying a class or all the classes are identified accurately. From the result, it is clear that all the classes were classified with almost precision after seeing the confusion matrix. It helps in assessing how currently selected classifier performs in each class.

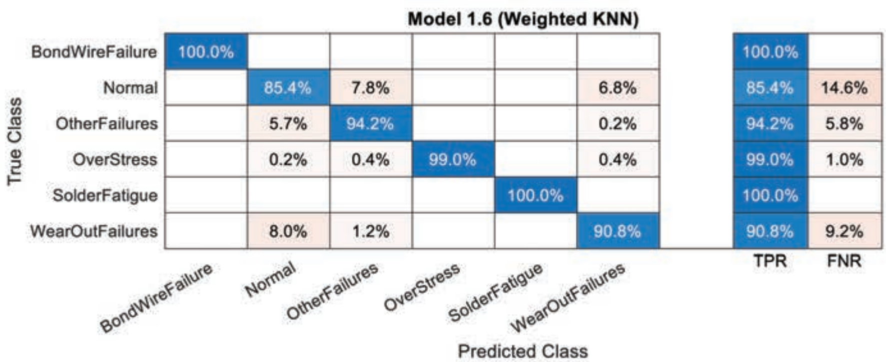
Figure 8.13 depicts the receiver operating characteristics by plotting the sensitivity (true positive rate) and 1-specificity (false positive rate) for different possible cut points in a training network. It is observed in Figure 8.10 that all the classes are close to the left and top border of the receiver operating characteristic (ROC) space making the training and testing process most accurate.

8.6 Failure mode effect classification analysis

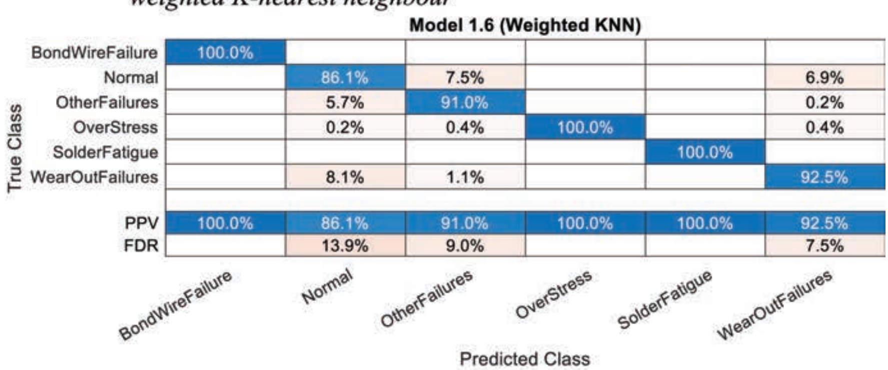
Both FMEA and FMMEA call for a criticality analysis to prioritize failure modes and mechanisms, respectively. Such prioritization allows for efficient allocation of resources for enabling and improving reliability of a system. One difficulty for prioritizing failure mechanisms for component level FMMEA is that the information necessary to make the decision is highly application dependent. This section will describe the traditional method for defining and estimating criticality and establish



(a) Truly and falsely classified features with weighted K-nearest neighbour

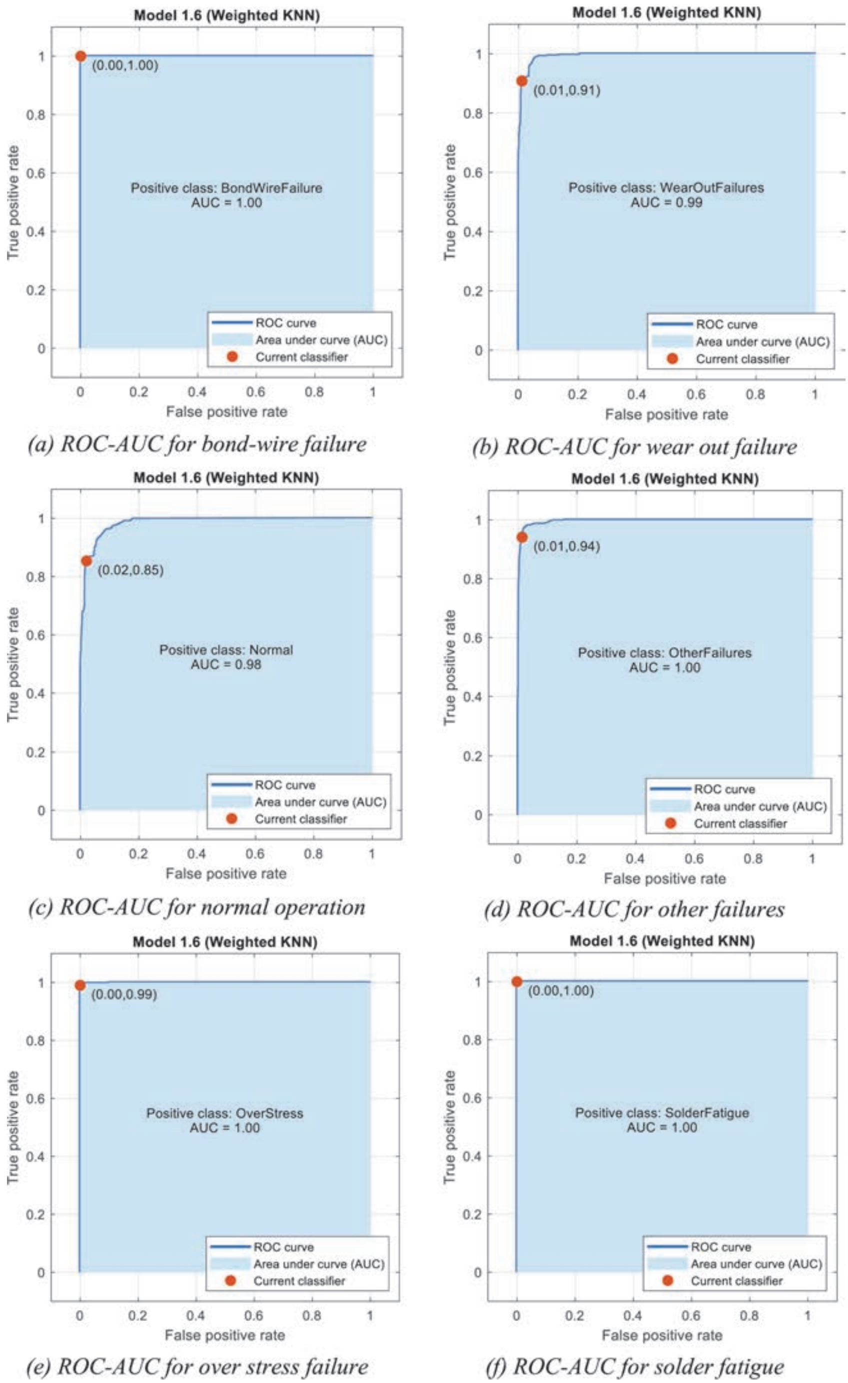


(b) True positive rate (TPR) and false negative rate (FNR) of features with weighted K-nearest neighbour



(c) Positive predictive values (PPV) and false discovery rate (FDR) of classification process

Figure 8.12 Confusion matrix for trained data



(a) ROC-AUC for bond-wire failure

(b) ROC-AUC for wear out failure

(c) ROC-AUC for normal operation

(d) ROC-AUC for other failures

(e) ROC-AUC for over stress failure

(f) ROC-AUC for solder fatigue

Figure 8.13 ROC for trained data

component-level information-based guidance for ranking failure mechanisms based on criticality.

8.6.1 Approaches for criticality analysis

Through JEP131B, JEDEC outlines three components for critically ranking failure modes: severity, occurrence and detection [48]. Each of the three categories is separately given a ranking from 1 to 10 based on the judgement of the team that is completing the analysis with 10 being the most severe, highest occurrence and most difficult to detect. These three metrics are then multiplied together create a risk priority number (RPN). Failure modes with higher RPNs are determined to be of more concern than those with lower RPNs. Corrective actions meant to be prioritized to lower the RPN of the highest failure modes. The RPN should be updated after corrective actions are taken and the design should be re-evaluated. As the rankings are based on the judgement of the team, RPN rankings should not be compared with another group's RPN ranking for the same or any other system.

The severity of a failure mode is dependent on the effects to the end user. First and foremost, severity should consider any potential to harm the users of a system. If there is potential to harm due to the effects from a certain mode (or mechanism), that mode (or mechanism) should be assigned a higher severity rating. The next considerations should include costs to the user and the system manufacturer. Costs take on a variety of forms but may include legal, warranty and returns, associated maintenance and brand reputation. Based on the judgement of the team a severity ranking should be given which takes into consideration these factors. It is evident that the severity of the failure is application dependent. Component level severity will be discussed subsequently.

The occurrence of a failure mode is how likely it is to occur. Considerations for occurrence should include environmental and loading conditions as well as system materials, geometries and part types. Using this information, it is possible to establish a probability of a mode or mechanism that can then be ranked according to the judgement of the FMEA development team. This information is also application dependent.

Finally, the detection metric for a failure mode is traditionally defined as the ability to detect a failure mode before shipping the product to the customer. Traditionally, in the electronics industry detection deals with the escape rate of any given test or screen. JEDEC suggests using inverse of the escape rate as one way of quantifying the detection of the mode. As the scope of the FMEA developed for a system level, individual power semiconductors are not assumed to be tested at the system integration level. Therefore, a different approach to detection will be taken in the subsequent discussion.

8.6.2 Approaches for severity analysis

Severity is determined by the effect to the end user. In the absence of this information, severity must be viewed in a different context. However, all Si power devices will be used in a larger circuit and thus have other electrical components nearby. For the purposes of a component level prioritization, severity will be determined by the potential of the failure to be catastrophic and effect nearby components.

With respect to Si power devices, overstress mechanisms which result in short circuit failure have the most potential to damage the system around them. Short circuits can create significant joule heating, which, if uncontrolled can damage nearby components and potentially start a fire thus increasing the associated costs of failure. The wear out mechanisms can be considered less severe because it is likely that only the Si power device fails, not harming other nearby components, and the system can potentially be repaired or replaced.

8.6.3 Occurrence

Occurrence is dependent on the application operating and environmental conditions. Certain failure mechanisms can only be expected to occur when specific stressors are present. For example, electrochemical migration is not a concern in an application where humidity is below the threshold for initiation. To calculate a value for an occurrence there are two approaches: one for wear out mechanisms and one for overstress mechanisms. For wear out mechanisms, one must identify a failure model which relates the stressors with the materials and geometries of the system, from this a time to failure or equivalent can occur giving an indication of the occurrence of the mechanism in the application. One example of this is the use of the Norris–Landsberg model for calculating fatigue of die attach. Failure models express time-to-failure, or equivalent, as a function of the stresses action on a system. Overstress failures are given a high priority with respect to occurrence as the stresses which are reasonably expected in the life cycle profile should be designed against. Assuming the proper design precautions have been taken, overstress mechanisms are unlikely to occur in the field and can only be quantified by identifying a probability that an overstress condition could occur. For example, the probability that a lightning strike causes a burnout of a device due to overcurrent can be a 'measure' of occurrence.

8.6.4 Detection

Detection in the traditional sense is determined by the ability to detect a failures, defects and non-conformities before it leaves manufacturing or assembly. For a component level discussion, this traditional definition is not applicable. Therefore, detection will be considered as the ability to detect a failure in operation, before the failure occurs. For an expected loading profile, overstress mechanisms can be avoided through proper selection of parts and appropriate de-rating. However, overstress failures may still occur due to random and unpredictable loading excursions such as lightning strikes or crashes.

Not all mechanisms are unpredictable as accumulating damage changes some observable and measurable parameters. The ability to monitor and predict failure is a field of study referred to as prognostics and health management (PHM). In PHM, in situ data is monitored and analysed for purposes of anomaly detection, fault classification and remaining useful life calculation. Wear out mechanism can be detectable and several groups have successfully implemented PHM for Si power devices [50, 86, 87]. Certain wear out mechanisms are more detectable than other wear out and overstress mechanisms depending on the feasibility and correlation with damage of electrical parameters associated with the mechanism. PHM techniques and methods are developing rapidly and reducing in cost and ease of implementation; however, the development and implementation of a PHM framework is not yet trivial and therefore it may only be cost efficient for certain critical components and applications. Additionally, competing failure mechanisms may convolute the measurement signals as they have the same or similar failure modes making it difficult to distinguish between the failure modes and take necessary corrective action.

8.7 Summary

This chapter developed a failure mode classification mechanism for condition monitoring of PV inverters. The developed algorithm performed signal preprocessing by DWT for noise removal, feature extraction and region of interest segmentation. The wavelet coefficients associated with the DWT were optimized by a novel approach based on HSA. Various types of features were extracted once the signal preprocessing is completed. The harmony search analysis proved to be very efficient in choosing the best wavelet coefficient depending upon the structure of the signal. The extracted features are assigned towards corresponding classes and randomly divided as training and test data for the purpose of evaluation of the classifier. K-NN is used to classify the fault conditions of PV inverters into normal and faulty status. A fivefold cross validation is performed to measure the performance of the classifier with input data. On validation, the developed approach depicted a training accuracy of 96.1 per cent. Further, criticality analysis is established from component-level information-based guidance for ranking failure mechanisms.

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Index

Reliability of Power Electronics Converters for Solar Photovoltaic Applications

The importance of power electronic converters for electricity grid equipment is increasing due to the growing distribution-level penetration of renewable energy sources. The performance of the converters mostly depends on interactions between sources, loads, and their state of operation. These devices must be operated with safety and stability under normal conditions, fault conditions, overloads, as well as different operation modes. Therefore, enhanced control strategies of power electronic converters are necessary to improve system stability.

This book for researchers and practitioners discusses enhanced control strategies, fault and failure mode classification mechanisms, and reliability analysis methods for PV modules, power electronic converters, and grid-connected PV systems, and thermal image-based monitoring. The technologies conveyed serve to improve the reliability and stability of power systems. Life calculation of converters, and case and reliability studies are included as well.

The international author team consists of researchers with a range of backgrounds from academia and industry.

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